

Problem

Solve Prob. 3.8 using mesh analysis.

Step-by-step solution

Step 1 of 4

Draw the following circuit diagram shown in Figure 1.

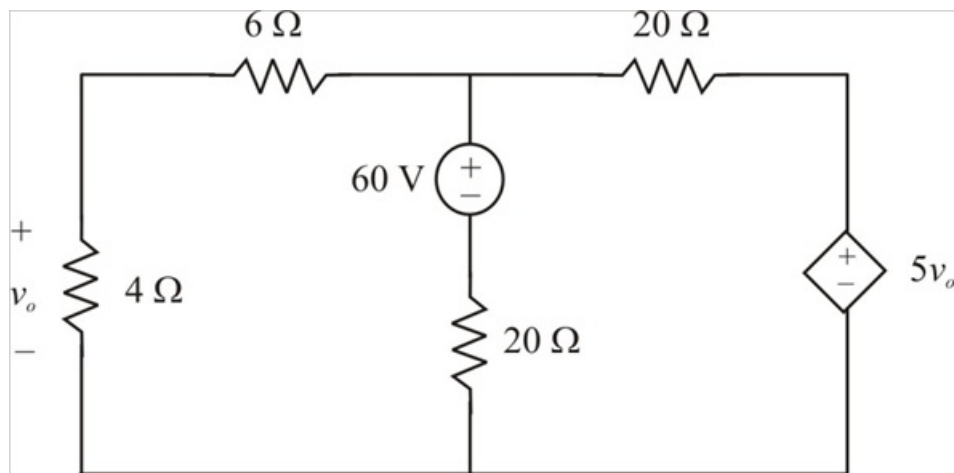


Figure 1

Step 2 of 4

The mesh currents are labeled to the circuit is shown in Figure 2.

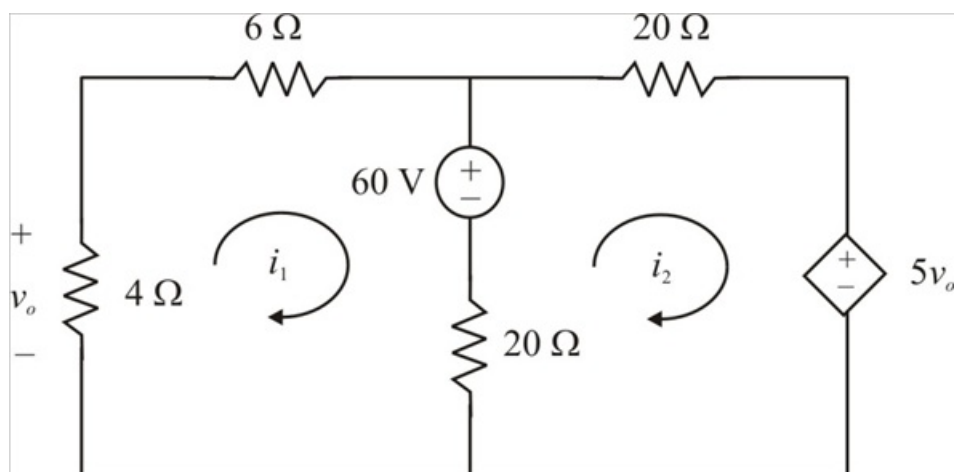


Figure 2

Step 3 of 4

Applying Kirchhoff's voltage law, the algebraic sum of all voltages around a closed path (or loop) is zero.

For mesh 1, Kirchhoff's voltage law gives,

$$4i_1 + 6i_1 + 60 + 20(i_1 - i_2) = 0$$

$$4i_1 + 6i_1 + 60 + 20i_1 - 20i_2 = 0$$

$$30i_1 - 20i_2 = -60$$

$$3i_1 - 2i_2 = -6$$

Therefore $3i_1 - 2i_2 = -6$ (1)

For mesh 2, Kirchhoff's voltage law gives,

$$20(i_2 - i_1) - 60 + 20i_2 + 5v_o = 0$$

$$20i_2 - 20i_1 - 60 + 20i_2 + 5(-4i_1) = 0$$

$$-40i_1 + 40i_2 = 60$$

$$-2i_1 + 2i_2 = 3$$

Therefore $-2i_1 + 2i_2 = 3$ (2)

Step 4 of 4

Adding equation (1) and equation (2), we get

$$\begin{array}{r} 3i_1 - 2i_2 = -6 \\ -2i_1 + 2i_2 = 3 \\ \hline i_1 = -3 \text{ A} \end{array}$$

Substituting the value of i_1 in equation (1), we get

$$3i_1 - 2i_2 = -6$$

$$(3)(-3) - 2i_2 = -6$$

$$2i_2 = -3$$

$$i_2 = -1.5 \text{ A}$$

The voltage v_o can be calculated as:

$$\begin{aligned} v_o &= -4i_1 \\ &= -4(-3) \\ &= 12 \text{ V} \end{aligned}$$

Therefore the voltage v_o is 12 V.